Retail Demand Decomposition & Classification Report

1. Project Overview

This project focuses on the deep decomposition and classification of retail demand patterns using 3,049 Walmart product time series from the M5 Forecasting competition. By mapping these patterns, we provide a strategic framework for inventory management and forecasting model selection.

2. Tech Stack & Libraries

- Python: Primary programming language.
- Pandas & NumPy: Data manipulation and numerical computation.
- Matplotlib & Seaborn: Data visualization.
- Statsmodels: Advanced statistical modeling (STL Decomposition, ADF, KPSS tests).

3. Data & Datasets

The analysis uses the following datasets located in the 'Datasets/' folder:
- calendar.csv: Contains date-specific information including holidays and SNAP events.
- sales_train_evaluation.csv: Daily sales for 3,049 items across 10 stores (aggregated for analysis).
- sell_prices.csv: Weekly price data per item and store.
The project aggregates these 30,490 series into 3,049 SKU-level series for higher-level pattern mapping.

4. Analytical Logic

A. SBC Classification (Syntetos-Boylan-Croston):
Items are classified into four categories based on Average Demand Interval (ADI) and variability (CV^2):
  - Smooth: High frequency, low variability.
  - Intermittent: Sparse demand, low variability.
  - Erratic: Frequent demand, high variability.
  - Lumpy: Sparse demand, high variability.

B. ABC/XYZ Analysis:
  - ABC: Revenue-based importance (Pareto 80/15/5 rule).
  - XYZ: Variability-based predictability (X=Stable, Z=Unpredictable).

C. STL Decomposition:

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Seasonal-Trend decomposition using LOESS (period=7) to extract underlying trends and weekly seasonality.

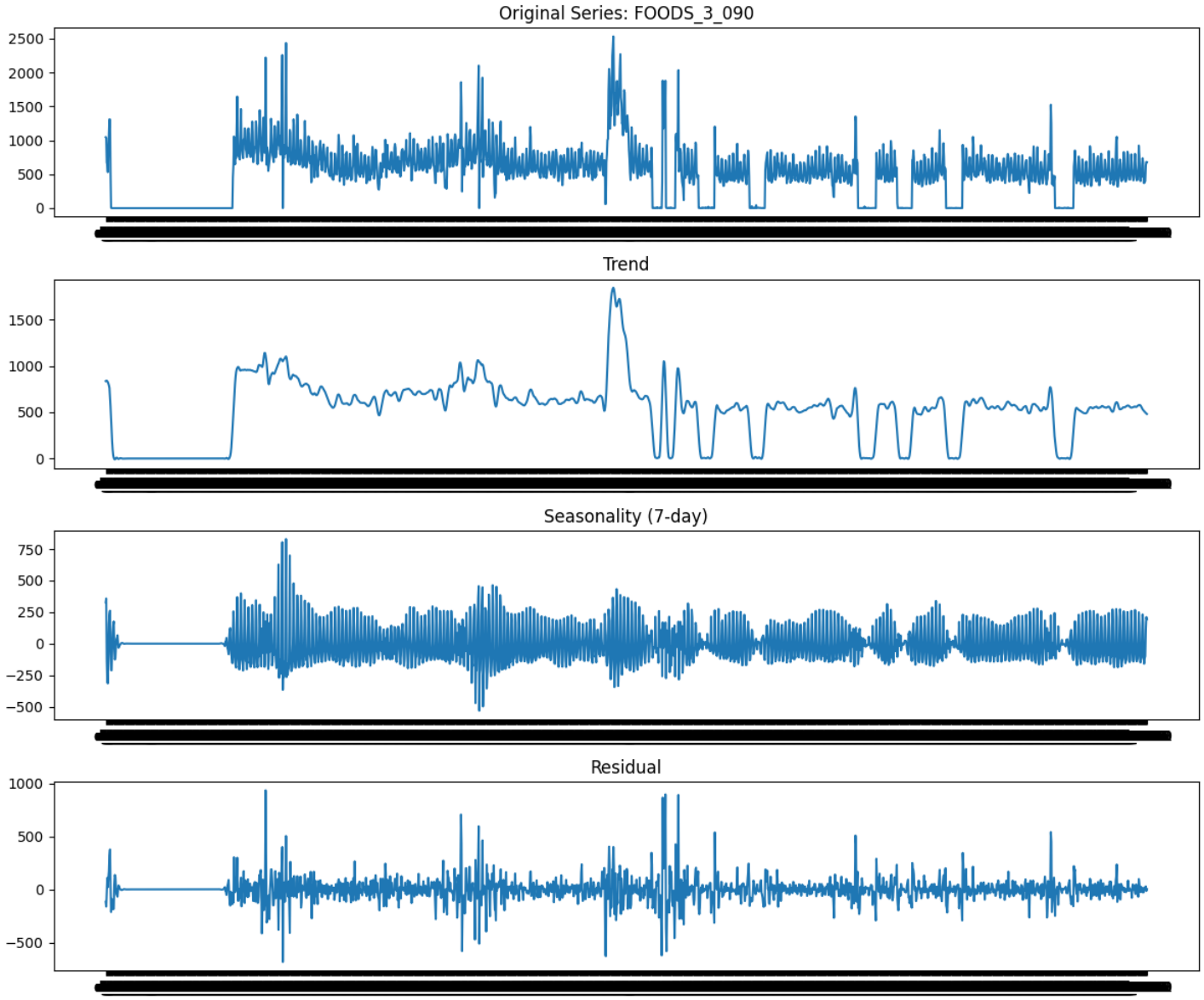
D. Stationarity Testing:
Applying ADF (Augmented Dickey-Fuller) and KPSS tests to determine if the series requires differencing.

5. Results & Outputs

The project generates several key files:
- demand_classification_results.csv: Full classification for all 3,049 SKUs.
- stationarity_results.csv: Statistical test results per item.
- category_sbc_distribution.csv: Proportional breakdown of demand patterns by product category (FOODS, HOBBIES, HOUSEHOLD).
- sample_decomposition.png: A visual breakdown of a typical 'Smooth' demand series.

Sample STL Decomposition Plot

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6. Strategic Recommendations

1. Smooth/A-X Items: Use automated EOQ models; these are high-value and predictable.
2. Intermittent/C Items: Use 'Pull' inventory systems with min-max levels.
3. Lumpy/Erratic Items: Deploy advanced ML models (e.g., LightGBM) to capture external drivers like promotions.